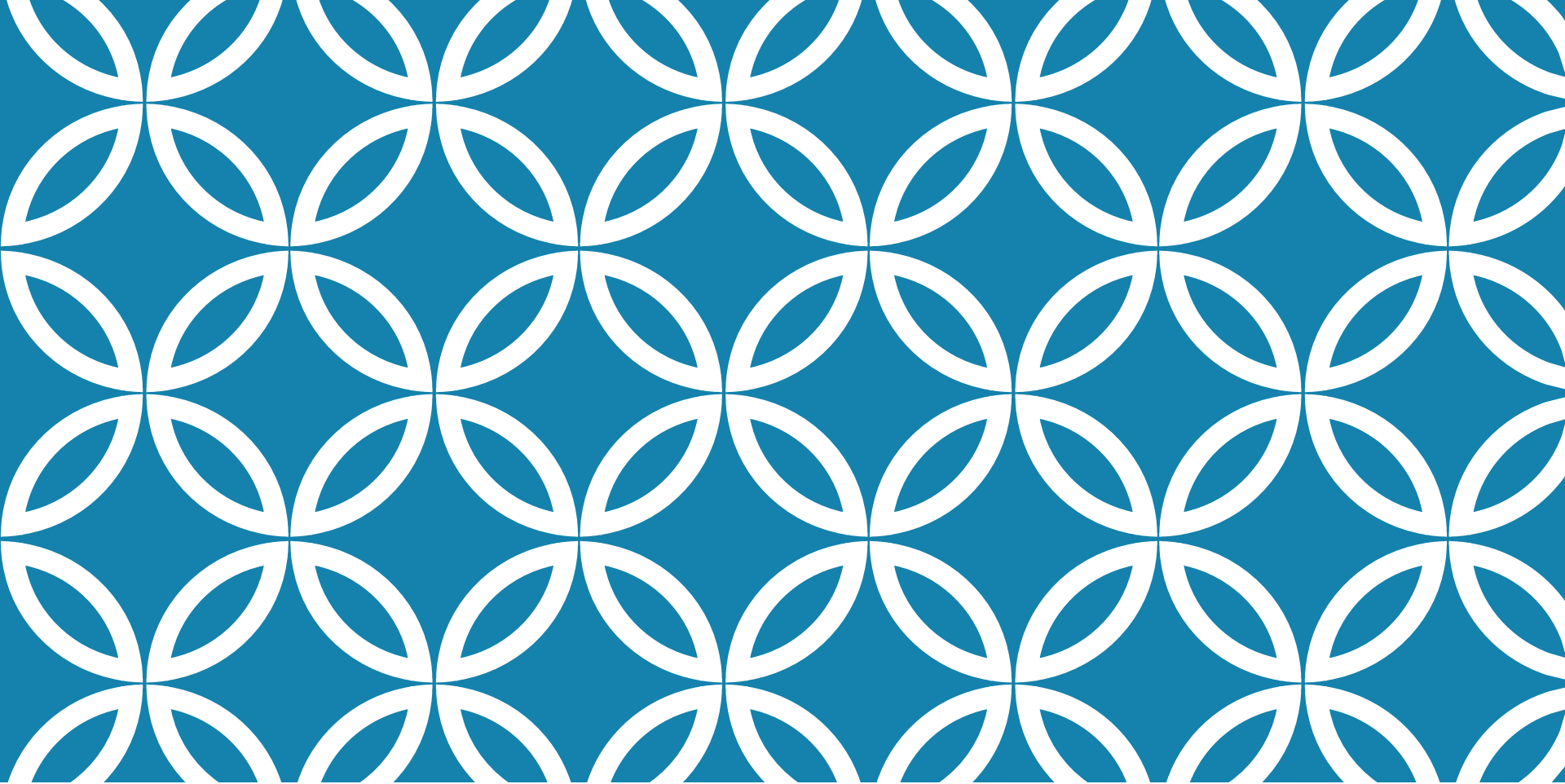




PRE-CALCULUS

NUMBER SYSTEMS

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2021 - 2022

Number systems

The **integers** are the set of all positive and negative whole numbers, and zero:

$$\dots, -3, -2, -1, 0, 1, 2, 3, 4, \dots$$

The set of integers is denoted by $\mathbb{Z}$. From the integers we can construct the **rational numbers**. These are all the ratios of integers. That is, any rational number r can be written as

$$r = \frac{m}{n} \text{ where } m, n \in \mathbb{Z}, n \neq 0.$$

The following are examples of rational numbers (elements of $\mathbb{Q}$):

$$\frac{1}{3} \quad \frac{22}{7} \quad \frac{-6}{7} \quad 3 = \frac{24}{8} = \frac{3}{1}$$

$$1.72 = \frac{172}{100} = \frac{43}{25} \quad 1 = \frac{1}{1}$$

Number systems

Some numbers cannot be written as $\frac{m}{n}$ for $m, n \in \mathbb{Z}$. These are the *irrational* numbers.

$$\sqrt{2} \quad \sqrt[3]{9} \quad \pi \quad e \quad \log_{10} 2$$

Combining the rational and irrational numbers gives us the set of *real numbers*, denoted $\mathbb{R}$. Every $x \in \mathbb{R}$ has a decimal expansion. For rationals, the decimal will start to repeat at some point. For example:

$$\frac{1}{3} = 0.33333 \dots = 0.\overline{3} \quad \frac{1}{7} = 0.\overline{142857}$$

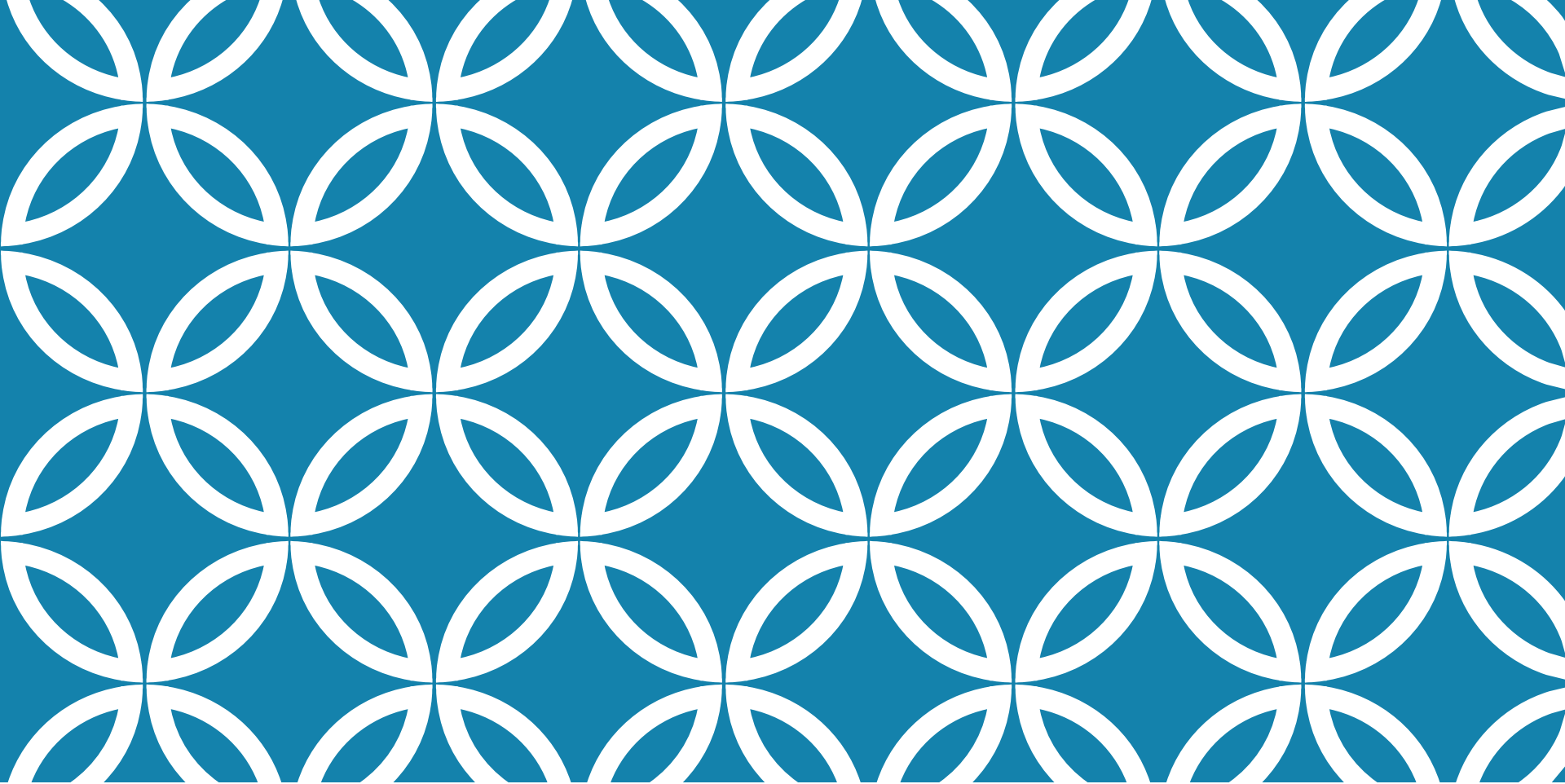
The real numbers

The real numbers are **totally ordered**. We can compare any two real numbers and say whether the first one is bigger than the second one, whether the second is bigger than the first, or whether they are equal.

The following are examples of true inequalities:

$$7 < 7.4 < 7.5 \quad -\pi < -3 \quad \sqrt{2} < 2$$

$$\sqrt{2} \leq 2 \quad 2 \leq 2 \quad -10 < \sqrt{100}$$



PRE-CALCULUS

SET NOTATION

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Set notation

A **set** is a collection of objects. If S is a set, we write $a \in S$ to say that a is an element of S . We can also write $a \notin S$ to mean that a is *not* an element of S .

Example: $3 \in \mathbb{Z}$ but $\pi \notin \mathbb{Z}$.

Example of set-builder notation:

$$\begin{aligned} A &= \{1, 2, 3, 4, 5, 6\} \\ &= \{x \in \mathbb{Z} \mid 0 < x < 7\} \end{aligned}$$

Another example of set notation

$$\begin{aligned}\{-2, -1, 0, 1, 2, 3\} &= \{x \in \mathbb{Z} \mid -2 \leq x \leq 3\} \\ &= \{x \in \mathbb{Z} \mid -3 < x \leq 3\} \\ &= \{x \in \mathbb{Z} \mid -3 < x < 4\} \\ &= \{x \in \mathbb{Z} \mid -2 \leq x < 4\}\end{aligned}$$

Intervals For $a, b \in \mathbb{R}$,

$$(a, b) = \{ x \in \mathbb{R} \mid a < x < b \}$$

whereas

$$[a, b] = \{ x \in \mathbb{R} \mid a \leq x \leq b \}.$$

TABLE 1.1 Types of intervals

	Notation	Set description	Type	Picture
Finite:	(a, b)	$\{x a < x < b\}$	Open	
	$[a, b]$	$\{x a \leq x \leq b\}$	Closed	
	$[a, b)$	$\{x a \leq x < b\}$	Half-open	
	$(a, b]$	$\{x a < x \leq b\}$	Half-open	
Infinite:	(a, ∞)	$\{x x > a\}$	Open	
	$[a, \infty)$	$\{x x \geq a\}$	Closed	
	$(-\infty, b)$	$\{x x < b\}$	Open	
	$(-\infty, b]$	$\{x x \leq b\}$	Closed	
	$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	Both open and closed	

Intersections and Unions

Intersection of two intervals:

$$\begin{aligned}(a, b) \cap (c, d) &= \\ \{x \in \mathbb{R} \mid x \in (a, b) \text{ AND } x \in (c, d)\} \\ &= \{x \in \mathbb{R} \mid a < x < b \text{ AND } c < x < d\}\end{aligned}$$

Union of two intervals:

$$\begin{aligned}(a, b) \cup (c, d) &= \\ \{x \in \mathbb{R} \mid x \in (a, b) \text{ OR } x \in (c, d)\} \\ &= \{x \in \mathbb{R} \mid a < x < b \text{ OR } c < x < d\}\end{aligned}$$

Examples: unions & intersections

Simplify and give your answer in interval notation.

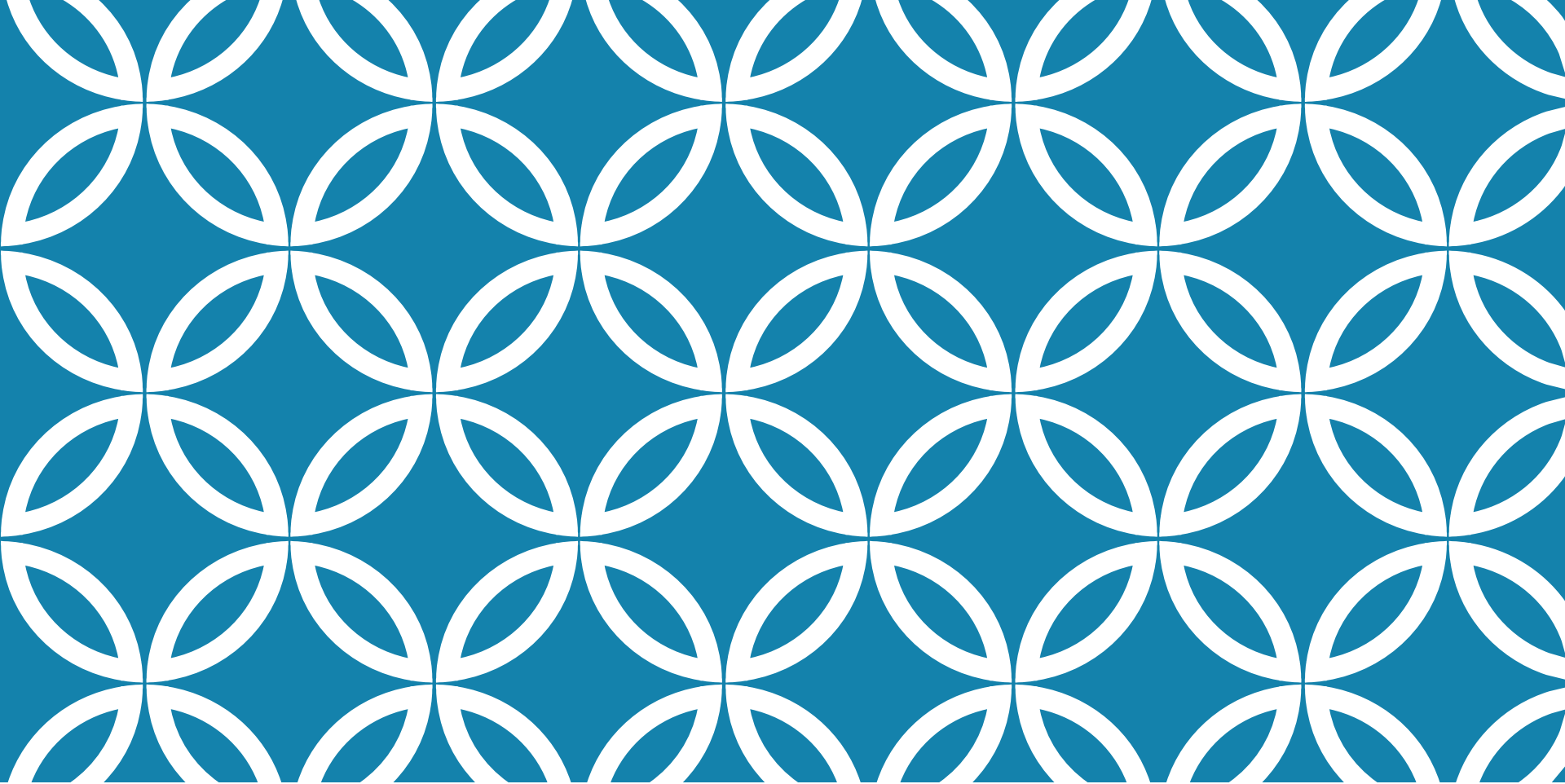
- ▶ $(1, 3) \cup (2, 4]$
- ▶ $[-4, \sqrt{2}) \cap [0, 1)$
- ▶ $[\sqrt{3}, 5) \cap (1.8, 7)$
- ▶ $[0, 4) \cap \left((-2, 1) \cup [3, 7) \right)$

Recall, $\sqrt{2} \approx 1.41 \dots$ and $\sqrt{3} \approx 1.73 \dots$

Examples: unions & intersection

Solutions:

- ▶ $(1, 3) \cup (2, 4] = (1, 4]$
- ▶ $[-4, \sqrt{2}) \cap [0, 1) = [0, 1)$
- ▶ $[\sqrt{3}, 5) \cap (1.8, 7) = (1.8, 5)$
- ▶ $[0, 4) \cap \left((-2, 1) \cup [3, 7) \right)$
 $= [0, 1) \cup [3, 4)$



PRE-CALCULUS

INEQUALITIES

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Understanding inequalities graphically

Consider the following functions:

$$f(x) = x^2 - 1$$

$$g(x) = (x - 1)^2$$

and the inequality $g(x) < f(x)$.

Also, notice the difference between solving for x in yesterday's tut question:

$$x^2 + 3x - 18 = 0$$

and

$$x^2 + 3x - 18 \geq 0$$

Rules for Inequalities

If a , b , and c are real numbers, then:

1. $a < b \Rightarrow a + c < b + c$

2. $a < b \Rightarrow a - c < b - c$

3. $a < b$ and $c > 0 \Rightarrow ac < bc$

4. $a < b$ and $c < 0 \Rightarrow bc < ac$

Special case: $a < b \Rightarrow -b < -a$

5. $a > 0 \Rightarrow \frac{1}{a} > 0$

6. If a and b are both positive or both negative, then $a < b \Rightarrow \frac{1}{b} < \frac{1}{a}$

Solving inequalities

We will often make use of a number line or a table to determine the sign of the function on particular intervals. We use critical values (where a function is 0 or undefined) to determine the intervals.

Examples:

1. Solve for x : $1 + x < 7x + 5$

2. Solve for x : $x^2 < 2x$

Solving $x^2 < 2x$

$$x^2 < 2x$$

$$\therefore x^2 - 2x < 0$$

$$\therefore x(x-2) < 0$$

$$\text{CVs: } x=0, x=2$$

$$\begin{array}{ccccccc} & + & & - & & + & \\ & \text{---} & & \text{---} & & \text{---} & \\ (x=-1) & 0 & (x=1) & 2 & (x=3) & & \end{array}$$

$$\therefore 0 < x < 2 \quad \text{or} \quad x \in (0, 2)$$

Solving inequalities continued

Find solutions to the following inequalities and write the solutions in interval notation.

1. $4 \leq 3x - 2 < 13$

2. $x^2 - 5x + 6 \leq 0$

3. $x^3 + 3x^2 > 4x$ (Don't divide by x !)

4. $\frac{x^2 - x - 6}{x^2 + x - 6} \leq 0$

5. $\frac{x^2 - 6}{x} \leq -1$ (Don't multiply by x !)

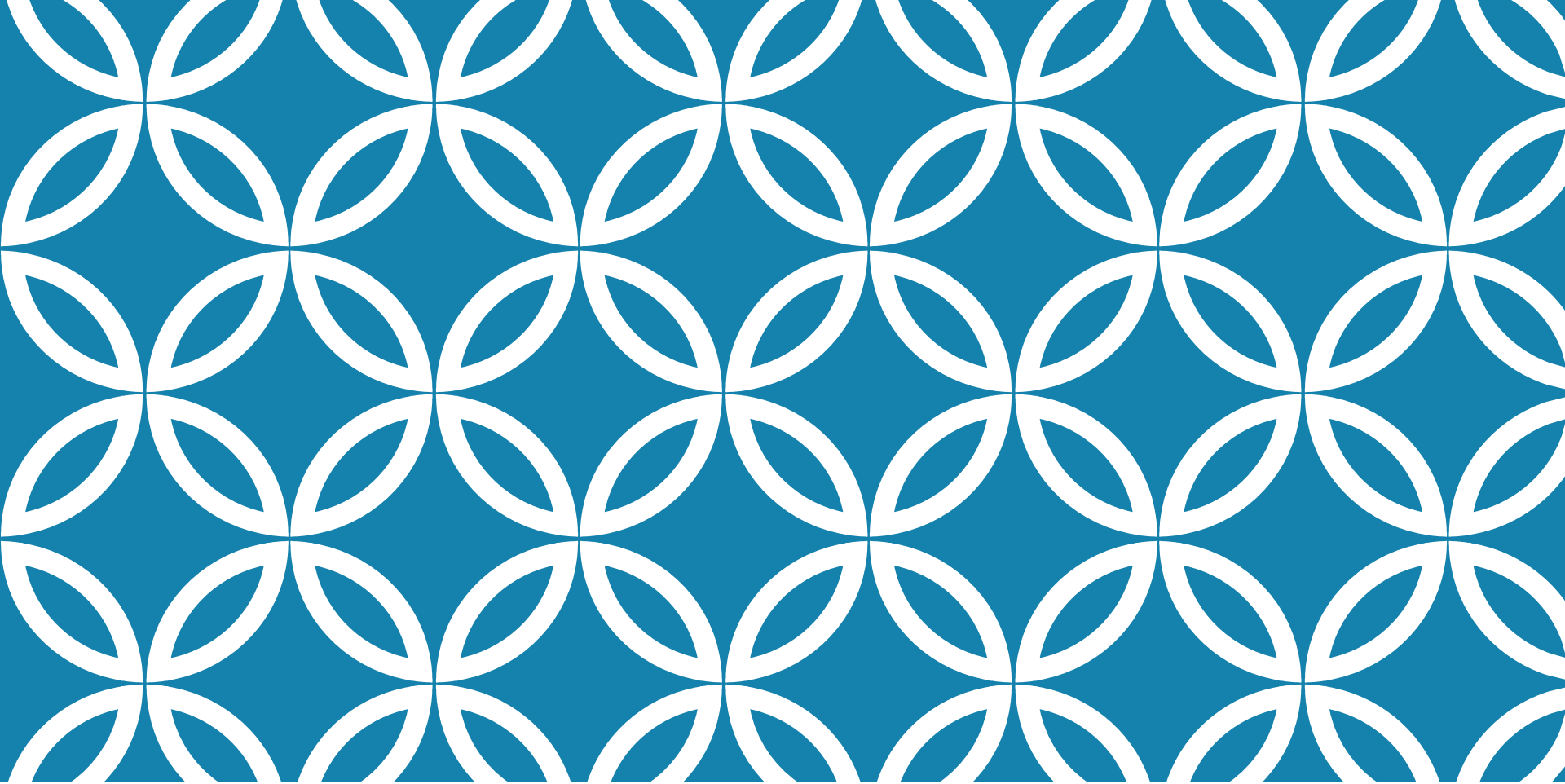
Solution to (5)

$$\frac{x^2-6}{x} + 1 \leq 0 \quad \therefore \frac{x^2+x-6}{x} \leq 0$$

$$\therefore \frac{(x+3)(x-2)}{x} \leq 0 \quad \text{CVs: } -3, 0, 2$$

	$x+3$	$x-2$	x	LHS
$(-\infty, -3)$	-	-	-	-
$(-3, 0)$	+	-	-	+
$(0, 2)$	+	-	+	-
$(2, \infty)$	+	+	+	+

$x \in (-\infty, -3] \cup (0, 2]$



PRE-CALCULUS

ABSOLUTE VALUE

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Absolute value

The **absolute value** of a number a , denoted by $|a|$ is the distance from a to 0 along the real line. A distance is always positive or equal to 0 so we have

$$|a| \geq 0 \text{ for all } a \in \mathbb{R}.$$

Examples

$$|3| = 3 \quad | - 3| = 3 \quad |0| = 0$$

$$|2 - \sqrt{3}| = 2 - \sqrt{3} \quad |3 - \pi| = \pi - 3$$

In general we have

$$|a| = a \quad \text{if } a \geq 0$$

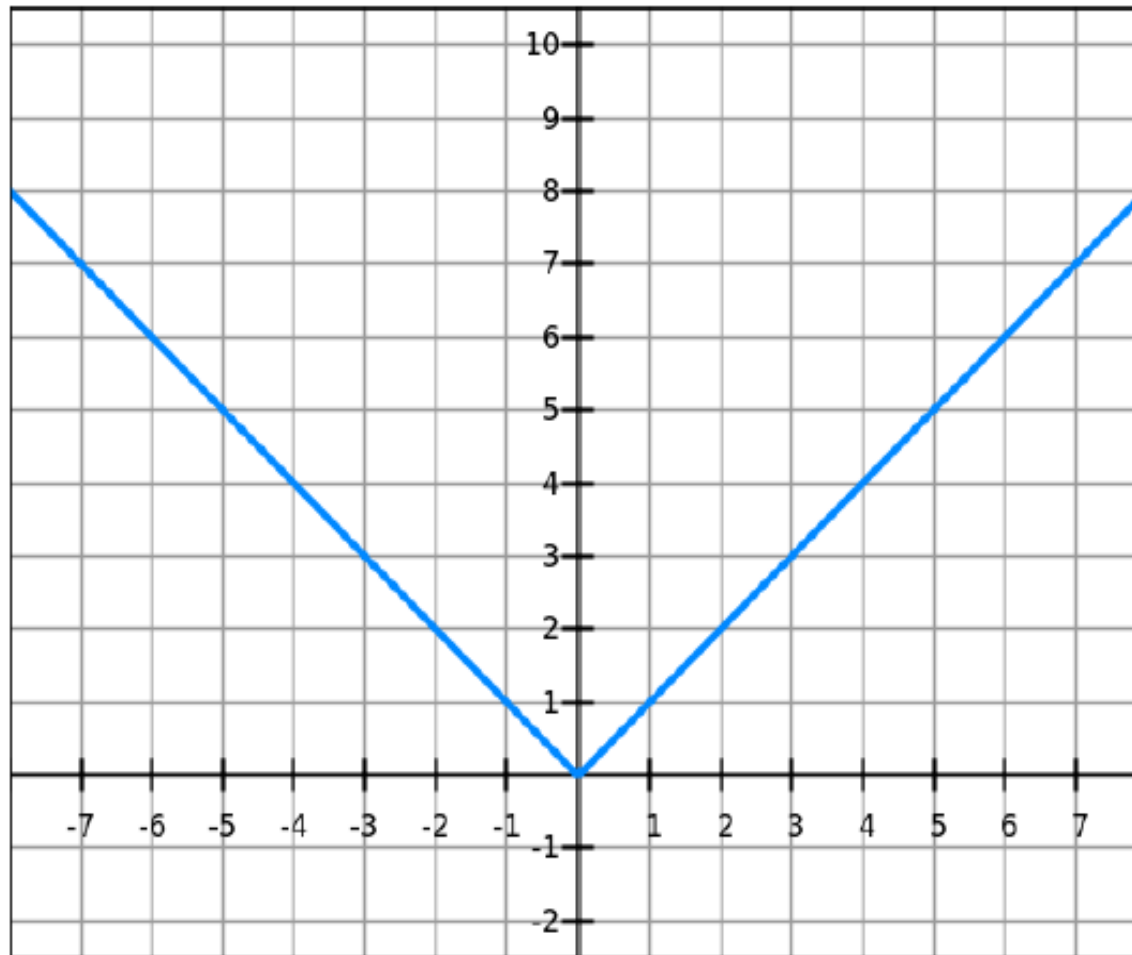
$$|a| = -a \quad \text{if } a < 0$$

We can write the absolute value function as a **piecewise defined function**.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

We can also replace the x above with something more complicated.

Sketching $y = |x|$:



If $f(x) = |x|$, calculate $f(-5)$, $f(4)$ and $f(0)$.

▶ $f(-5) = -(-5) = 5$

▶ $f(4) = 4$

▶ $f(0) = 0$

Note: for any $a \in \mathbb{R}$, $|a| = \sqrt{a^2}$

Example

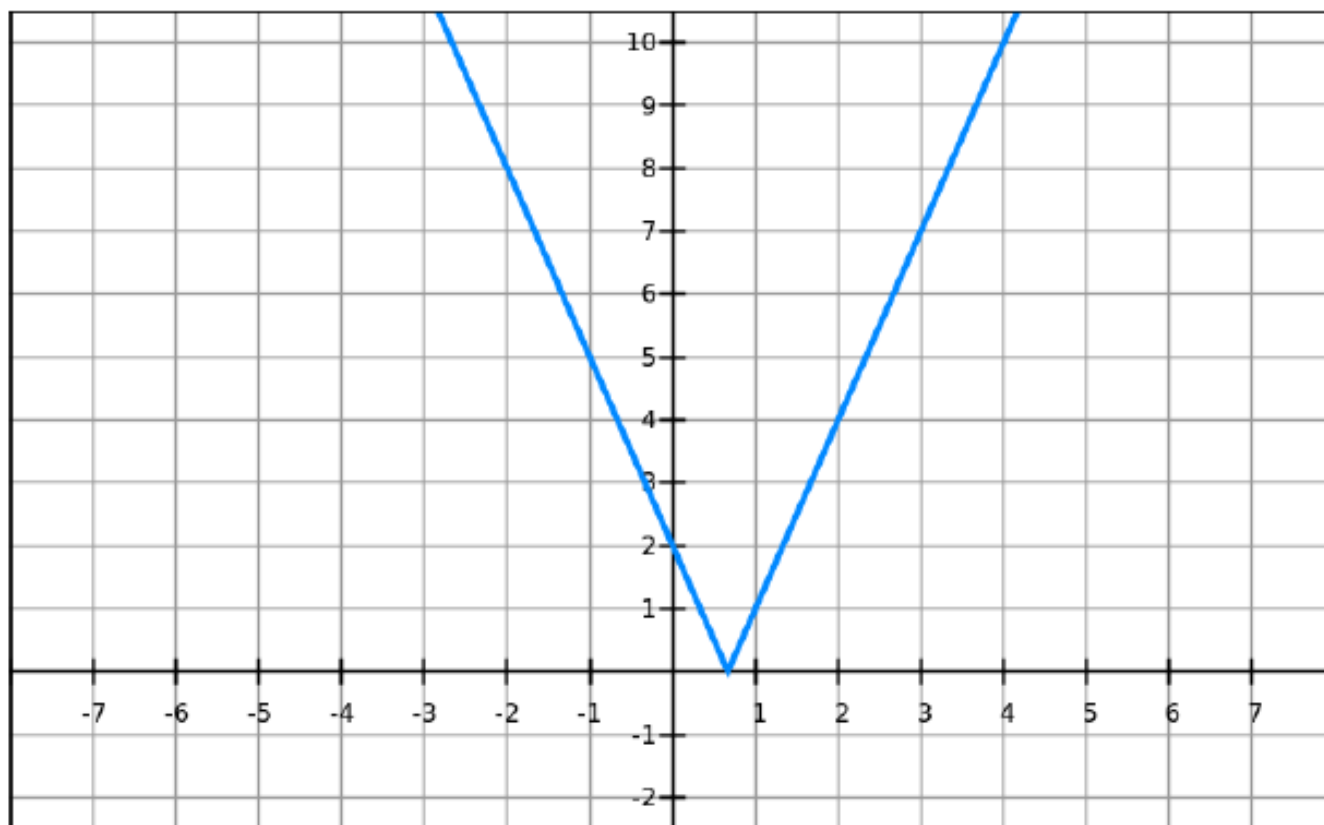
Write $|3x - 2|$ without using the absolute value symbol.

$$|3x - 2| = \begin{cases} 3x - 2 & \text{if } 3x - 2 \geq 0 \\ -(3x - 2) & \text{if } 3x - 2 < 0 \end{cases}$$

Hence

$$|3x - 2| = \begin{cases} 3x - 2 & \text{if } x \geq \frac{2}{3} \\ 2 - 3x & \text{if } x < \frac{2}{3} \end{cases}$$

Below is a sketch of $y = |3x - 2|$. Note how the function changes at $x = \frac{2}{3}$.



Properties of Absolute Values

Suppose $a, b \in \mathbb{R}$ and $n \in \mathbb{Z}$. Then

1. $|ab| = |a||b|$
2. $|\frac{a}{b}| = \frac{|a|}{|b|}$ ($b \neq 0$)
3. $|a^n| = |a|^n$

Let $a > 0$. Then

4. $|x| = a$ if and only if $x = a$ or $x = -a$.
5. $|x| < a$ if and only if $-a < x < a$.
6. $|x| > a$ if and only if $x > a$ or $x < -a$.

Example: Solve $|2x - 5| = 3$.

Solving absolute value inequalities:

- ▶ Solve: $|x - 5| < 2$.
- ▶ Solve: $|3x + 2| \geq 4$.

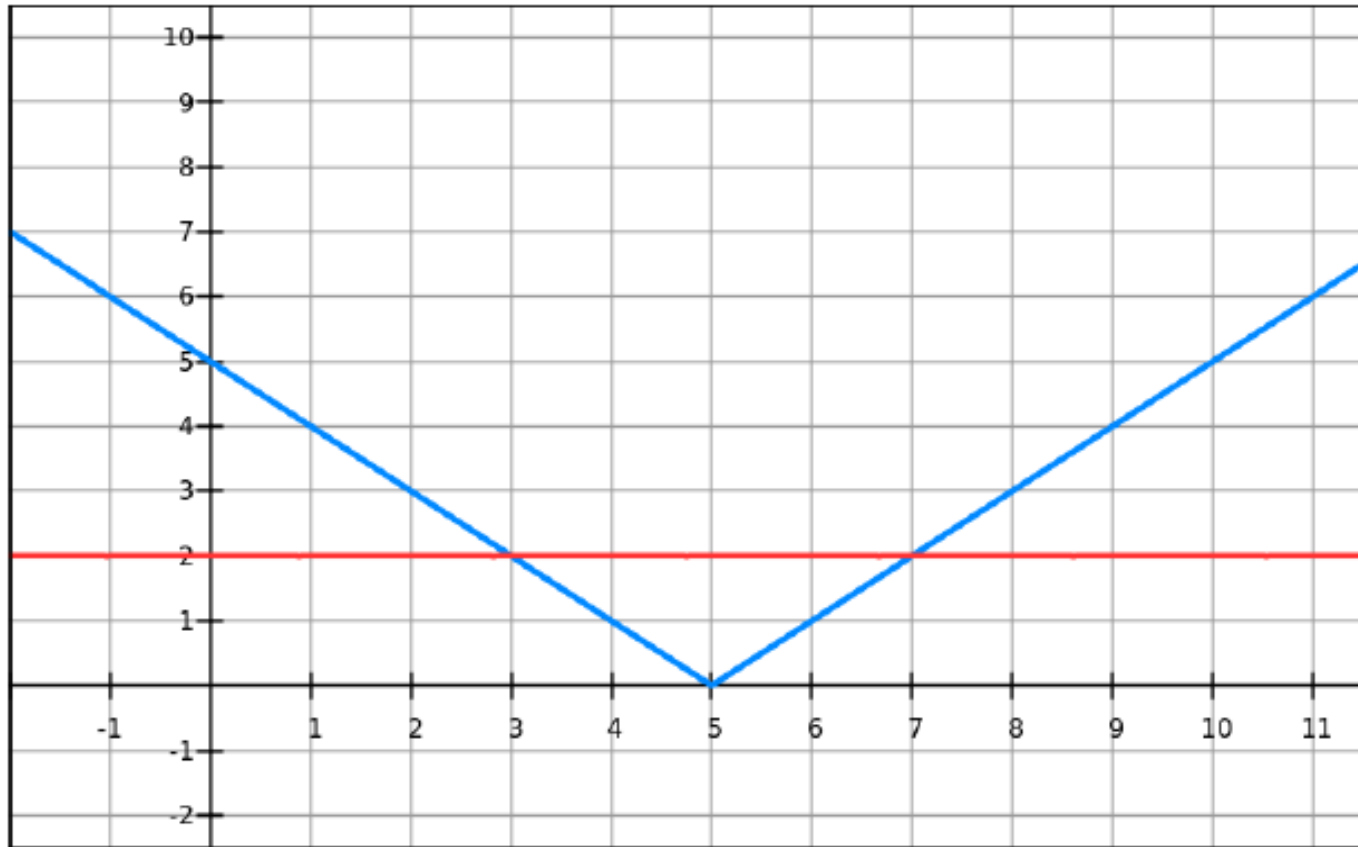
To solve the first inequality above we must solve

$$-2 < x - 5 < 2$$

For the second inequality we have

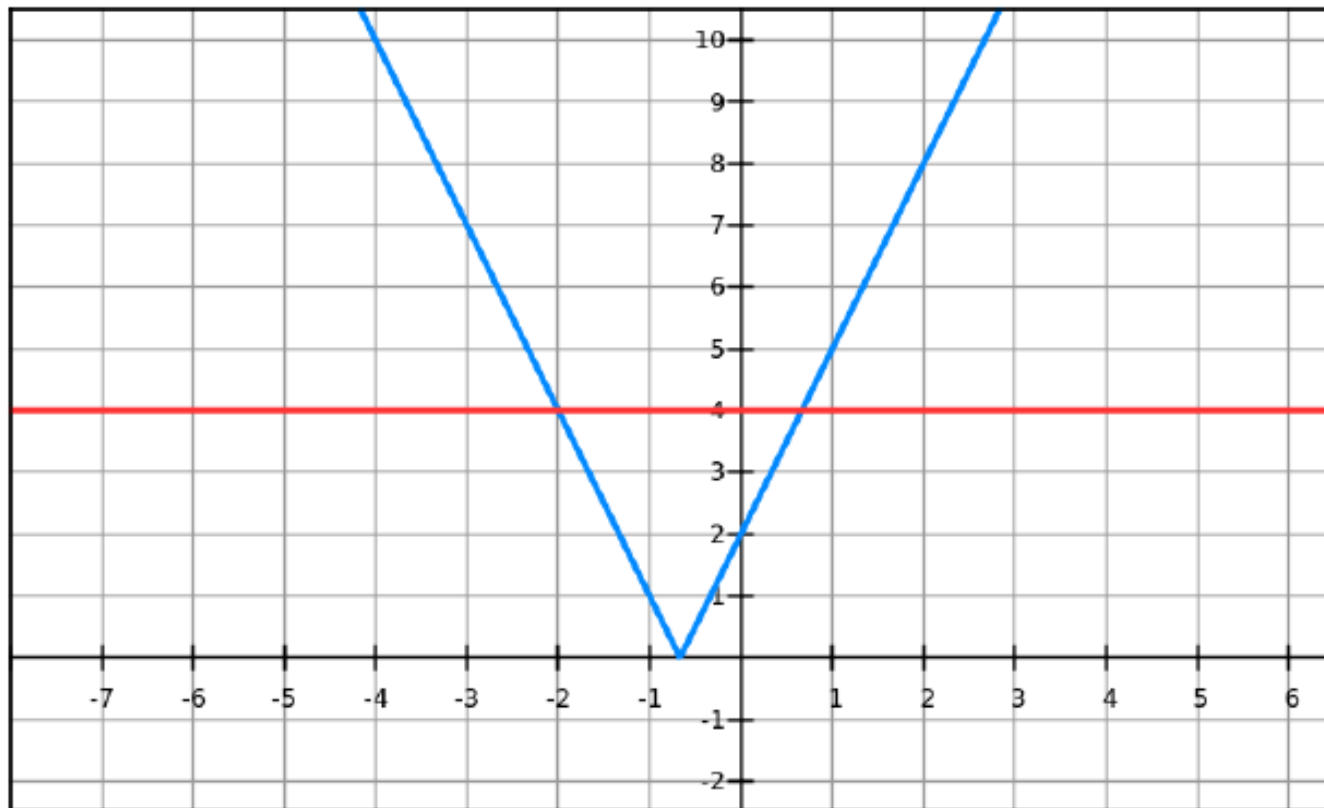
$$3x + 2 \geq 4 \quad \text{OR} \quad 3x + 2 \leq -4$$

Solving $|x - 5| < 2$ gives $x \in (3, 7)$.



Solving $|3x + 2| \geq 4$ gives

$$x \in (-\infty, -2] \cup [\frac{2}{3}, \infty)$$



Warm up

Let $a > 0$. Then

4. $|f(x)| = a$ if and only if $f(x) = a$ or $f(x) = -a$.

5. $|f(x)| < a$ if and only if $-a < f(x) < a$.

6. $|f(x)| > a$ if and only if $f(x) > a$ or $f(x) < -a$.

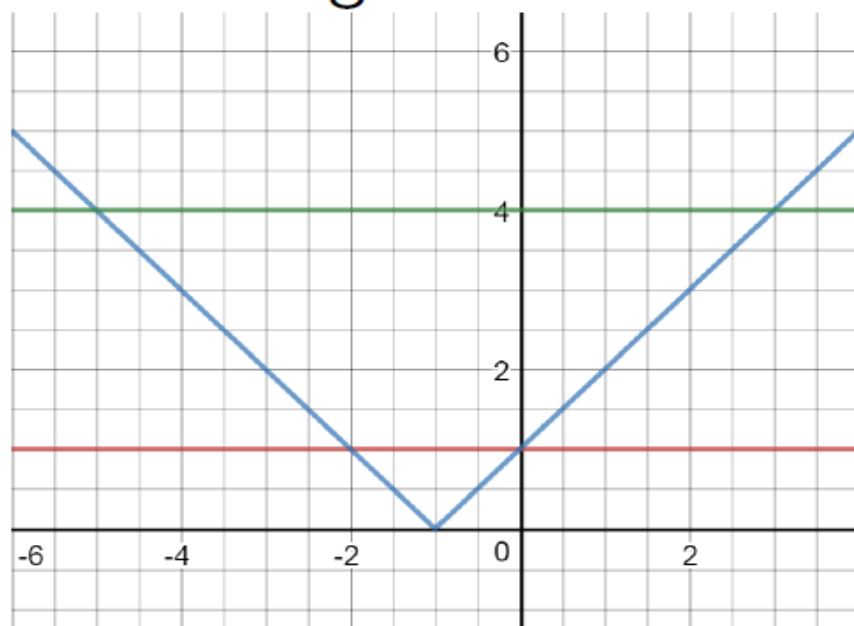
Now solve:

$$1 < |x + 1| < 4$$

The solution is the values of x that satisfy:

$$1 < |x + 1| \quad \text{AND} \quad |x + 1| < 4$$

Therefore $x \in (-5, -2) \cup (0, 3)$. This algebraic solution agrees with the sketch:



The triangle inequality

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

How do we prove this?

First observe that $-|a| \leq a$ and $a \leq |a|$.

Similarly, $-|b| \leq b$ and $b \leq |b|$.

Applications:

Example: If $|x - 4| < 0.1$ and $|y - 7| < 0.2$, use the Triangle Inequality to estimate $|(x + y) - 11|$.

Exercise: show that if $|x + 3| < \frac{1}{2}$, then $|4x + 13| < 3$.

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Website: <http://andrewcraigmaths.wordpress.com>

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